PHY202 Practice Exam 7 on chapter(s): 27, 28

1. $\theta = 22.8^\circ \because d \sin\theta = m\lambda$

2. $E = 1.53 \times 10^{15}\,\mathrm{J} \because E_0 = mc^2$

3. $f = 6.93 \times 10^{8}\,\mathrm{Hz} \because \lambda_{obs} = \lambda_s \sqrt{\dfrac{1+\dfrac{u}{c}}{1-\dfrac{u}{c}}}, c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}} = \dfrac{E}{B} = f\lambda$

4. $\lambda = 8.22 \times 10^{-7}\,\mathrm{m} \because d \sin\theta = m\lambda$

5. $\Delta t = 3.58 \times 10^{-7}\,\mathrm{s} \because \gamma = \left(\sqrt{1 - \dfrac{v^2}{c^2}}\right)^{-1}, \Delta t = \gamma \Delta t_0$

6. $\lambda_{obs} = 1.44\,\mathrm{m} \because \lambda_{obs} = \lambda_s \sqrt{\dfrac{1+\dfrac{u}{c}}{1-\dfrac{u}{c}}}$

7. $L = 7.10\,\mathrm{ly} \because L = L_0 \gamma^{-1}$

8. $\theta = 0.252\,\mathrm{arcseconds} \because \theta = 1.22\dfrac{\lambda}{D}$

9. $\theta = 45.8^\circ \because I = I_0 \cos^2\theta$

10. $D = 8.34 \times 10^{-6}\,\mathrm{m} \because D \sin\theta = m\lambda$

11. $x = 3.04 \times 10^{3}\,\mathrm{m} \because \theta = 1.22\dfrac{\lambda}{D}, \Delta\theta = \dfrac{\Delta s}{r}, \Delta s \approx 0.53\,\mathrm{m}$ for small angles

12. $t = 7.77 \times 10^{-8}\,\mathrm{m} \because 2t = \left(m + \dfrac{1}{2}\right)\lambda_n, n_{\mathrm{air}} < n_{\mathrm{oil}} > n_{\mathrm{water}}$